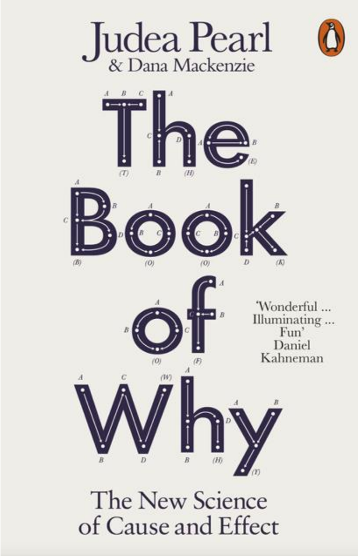
Causality

Data Science

Causality

- causal inference
- confounding
- DAG
- do-formalism
- counterfactuals
- inverse propensity score weighting for ML

Why Causality Matters



story behind the data as important as the data itself

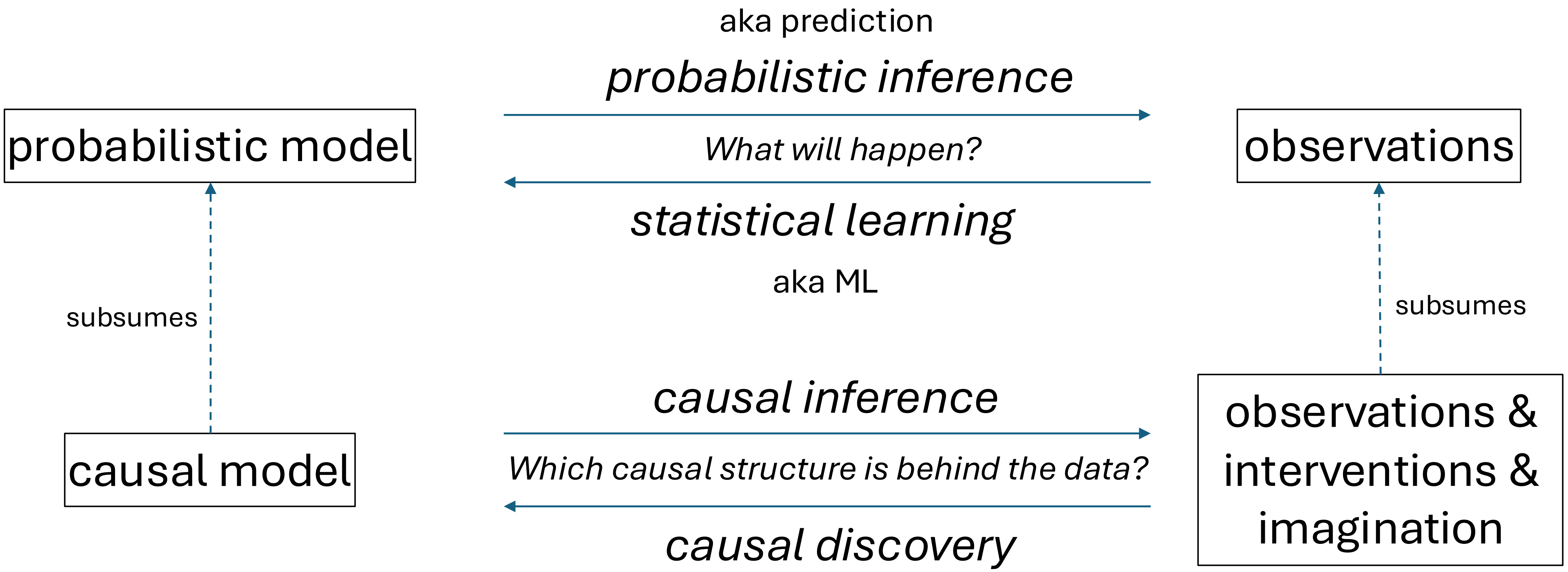
Why are the statistical dependencies as observed?

→ data-generating process mostly governed by causal dependencies

but language of algebra symmetric

no way to tell that a storm causes barometer to go down and not the other way around → need for asymmetric mathematical language

Probabilistic and Causal Models



from *Elements of Causal Inference* (Peters, Janzing, Schölkopf)

Probabilistic models describe patterns in observed data.
Causal models describe cause-and-effect relationships and also allow for interventions and counterfactuals.

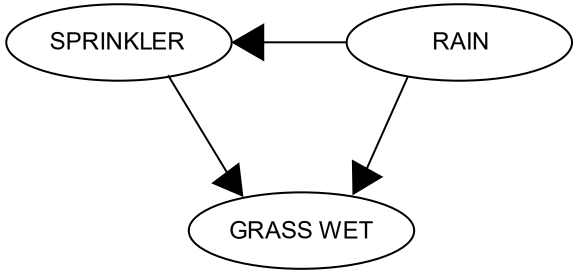
Probabilistic Graphical Model

Directed Acyclic Graph (DAG): random variables (nodes) and their (conditional) dependencies (edges)

Bayesian network: DAG with updates on new evidence according to Bayes' rule (belief propagation)

- message from parent to child: update using conditional probabilities
- message from child to parent: update by multiplication with likelihood ratio

	SPRINKLER	
RAIN	T	F
F	0.4	0.6
T	0.01	0.99



RAIN	
T	F
0.2	0.8

		GRASS WET	
SPRINKLER	RAIN	T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

from wikipedia

Markov Property

Each variable in a DAG is independent of its non-descendants conditional on its parents.

chain rule:
$A \rightarrow B \rightarrow C$: $\quad P(A, B, C) = P(A)P(B|A)P(C|B)$

→ compact representation of conditional probability table

→ without DAG, one would need to model the entire joint distribution $P(A, B, C)$, what becomes extremely large for many variables

Conditional Independencies

DAG junctions:

- chains: $A \rightarrow B \rightarrow C$ (A and C conditionally independent given B)
- forks: $A \leftarrow B \rightarrow C$ (A and C conditionally independent given B)
- colliders: $A \rightarrow B \leftarrow C$ (A and C conditionally dependent given B)

d-separation (d means directional): conditional independencies in data corresponding to chains, forks, and colliders in a graph

→ allows for model testing

Toward Causal Diagrams

load graph with causal assumptions (model)

→ arrows give direction of cause-effect relationships

causal Markov property: Each variable in a causal graph is probabilistically independent of its non-effects conditional on its direct causes.

causal models (represented by graphs) allow to answer interventional and counterfactual (fictional) queries

often even without conducting experiments

The Ladder of Causation

III	*imagining*	**counterfactuals**	What if I had done ...? Why?
II	*doing*	**intervention**	What if I do ...? How?
I	*seeing*	**association**	What if I see ...? → Realm of ML

from *The Book of Why* (Pearl, MacKenzie)

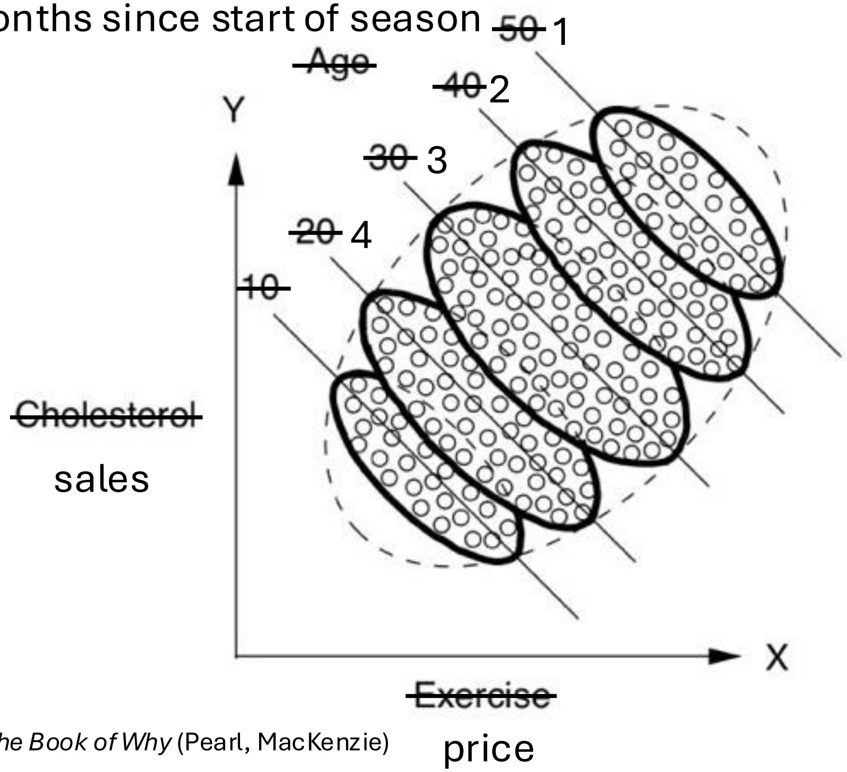
Example:

III What would have happened if I hadn't sent the customer a voucher?
II What happens when I send vouchers?
I What do I see? Shoppers with vouchers spend more.

Simpson’s Paradox

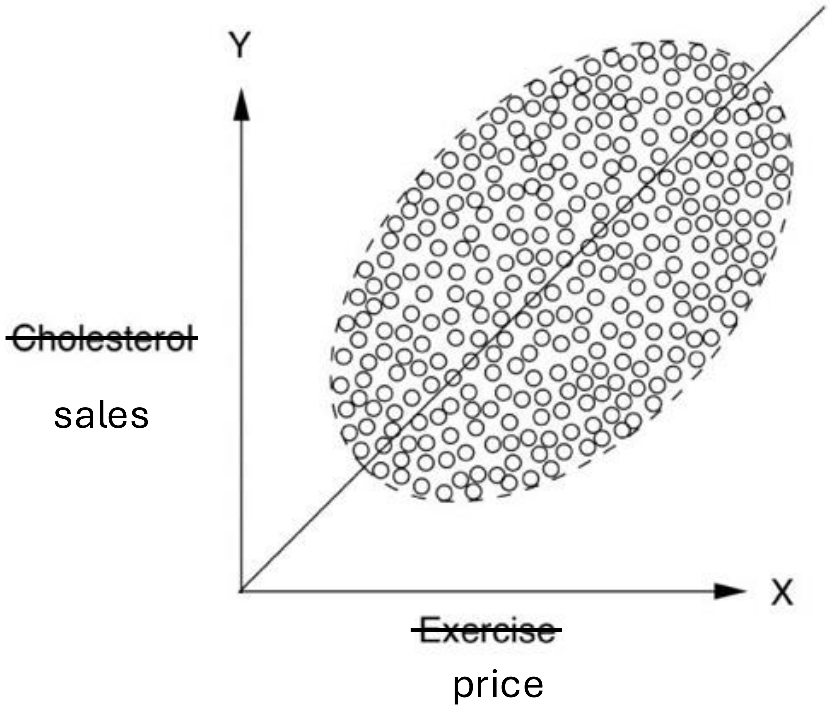
example: monthly sales of a fashion article in different shops during winter season
Lower price yields higher sales for each month individually, but lower sales overall?

months since start of season



from *The Book of Why* (Pearl, MacKenzie)

all data together ignoring monthly information



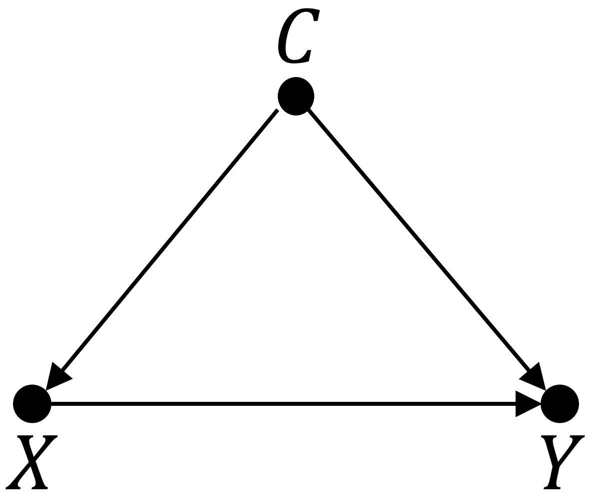
month is a hidden influencing factor on both variables:

sales
↑
month
↓
price

Confounding

causal effect of X on Y confounded by common cause C:

$$X \leftarrow C \rightarrow Y$$



→ spurious correlation between X and Y

(overlaying potential true causal effect of X on Y)

common cause principle:

every correlation either due to a direct causal effect linking the correlated entities or brought about by a third factor (confounder)

An observed correlation can be completely misleading if key influencing factors are not taken into account. →Is there a variable C that influences both X and Y?

Example: Pricing in Retail

price setting for a product can be considered a demand shaping method

the idea is a causal effect: lower price leads to higher demand

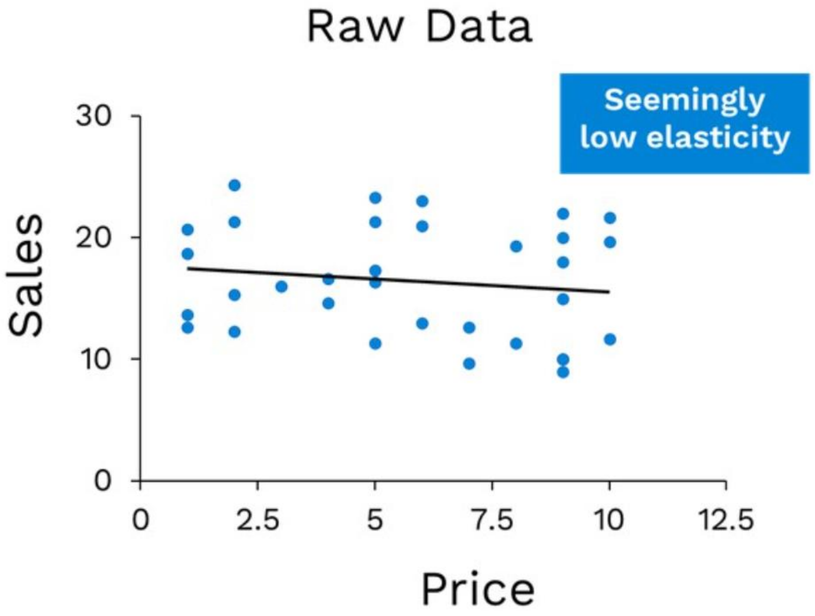
→ by estimating this causal effect, one can find an optimal price according to a given policy (like maximizing profit)

problem: confounders influencing both pricing in past (obverved) data and demand, e.g., lowering prices on weekends only for grocery or lowering prices toward end of a season for fashion (most sales at beginning of season)

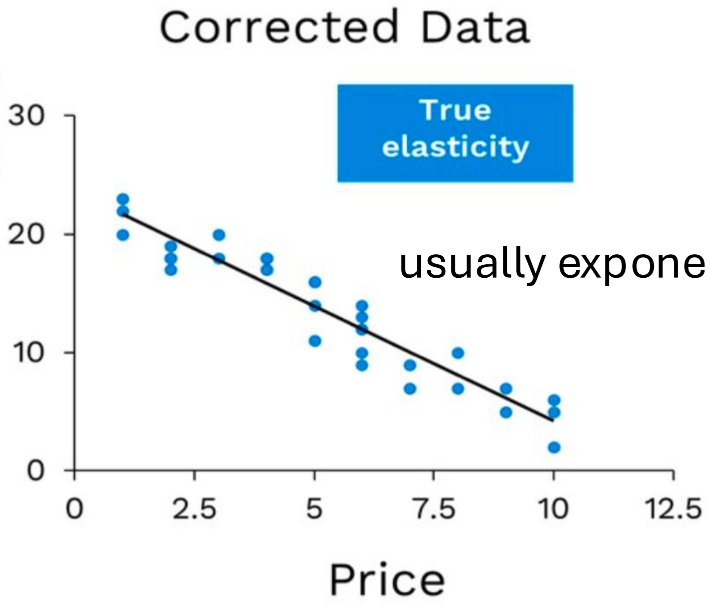
Demand Shaping with Causal Inference

dynamic pricing:
influencing demand of different products by price setting

confounded effect:

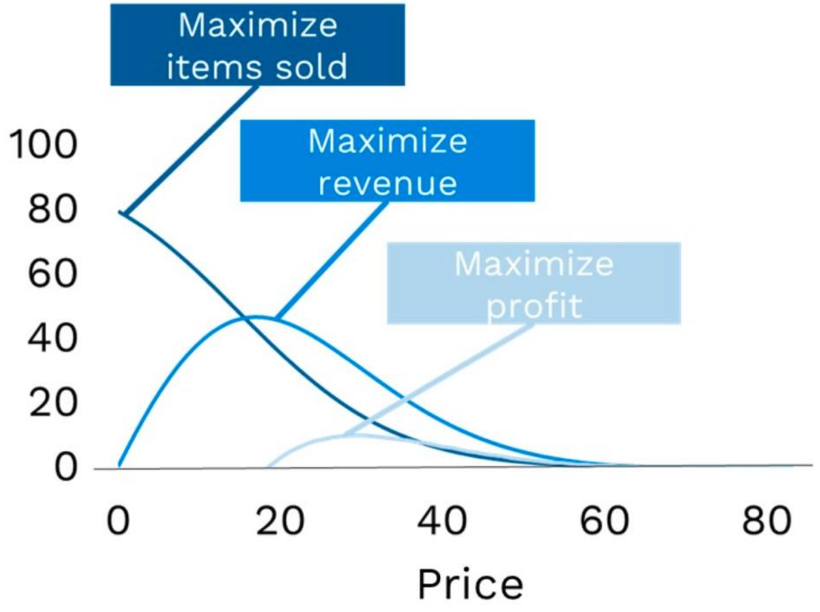


after de-confounding:



usually exponential

use for pricing policies:



Interventions vs Observations

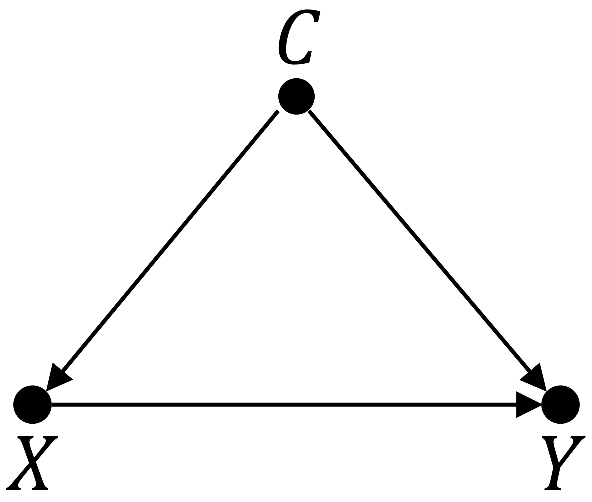
goal: prediction of (average) causal effect of (yet untried) actions (interventions)

two ways:

- intervene via randomized controlled trials (RCT): set action randomly and check effect → physically disassociate cause from confounders
- causal model: emulating interventions by smart calculations → predict effects of potential interventions from observational studies only, i.e., without experiment

Solution for Average Causal Effects

causal effect of X on Y confounded by common cause C



solution for computation of average causal effect:

adjust for C (if there are measurements of C), i.e., stratification of data in terms of C

$$\sum_c P(Y|X, C = c)\, P(C = c)$$

fashion example: adjust for months-in-season groups

Colliders and Mediators

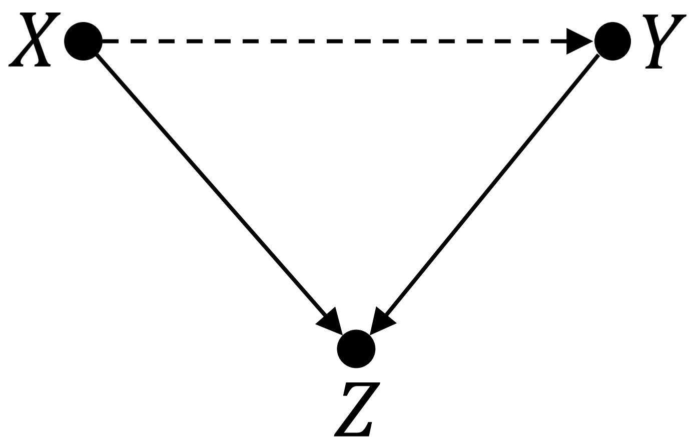
unwanted effects of adjusting for third variable on causal effect between X and Y:

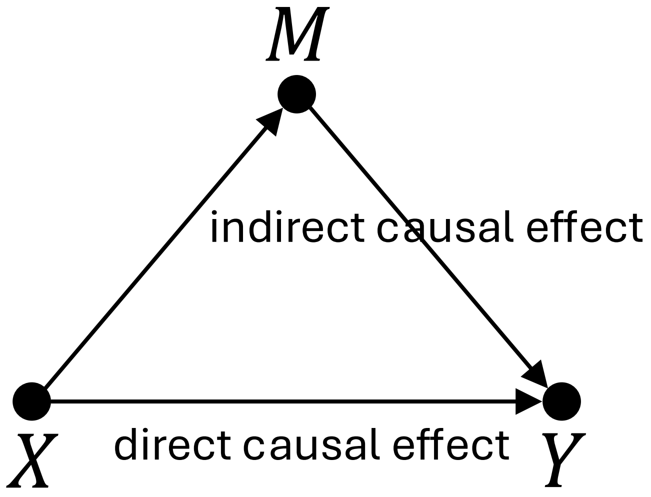
collider Z

- correlation without causation (exception to common cause principle)
- selection bias

mediator M (mechanism of effects)

→need for counterfactual description





Backdoor Criterion

to stop information flow from X to Y (d-separation):

- $X \rightarrow Z \rightarrow Y$ → adjust for Z
- $X \leftarrow Z \rightarrow Y$ → adjust for Z
- $X \rightarrow Z \leftarrow Y$ → do **not** adjust for Z

back-door path: any path from X to Y starting with an arrow pointing into X

de-confounding X and Y means blocking all back-door paths
(which need to be identified before)

do-Calculus

effect of intervention represented by $P(Y|do(X))$

goal: calculate $P(Y|do(X))$ in terms of data such as $P(Y|X, A, B, \ldots)$

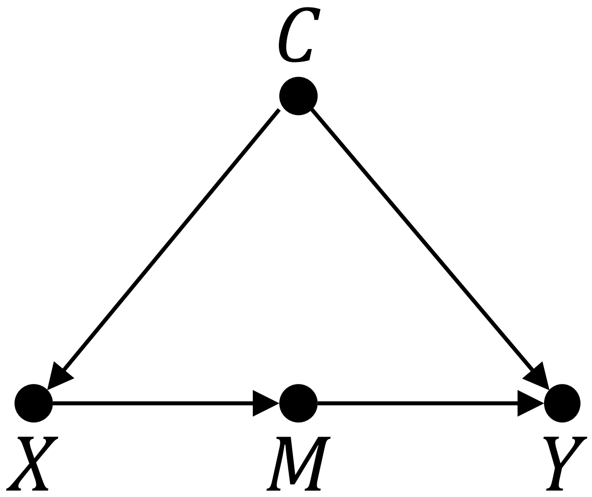
no do-operator remaining → can be calculated from observational data

$do(X)$ means removing all arrows of a causal diagram going into X

definition of confounding: $P(Y|X) \neq P(Y|do(X))$

back-door adjustment: $P(Y|do(X)) = \sum_c P(Y|X, C = c)\, P(C = c)$

Frontdoor Criterion



unobservable confounder → no back-door adjustment

front-door adjustment by means of observable mediator M:

$$P(Y|do(X)) = \sum_m P(M = m|X) \sum_x P(Y|X = x, M = m)\, P(X = x)$$

adjust for two variables: M and X

derived from simple axioms of do-calculus:

- observation of W irrelevant to Y: $P(Y|do(X), Z, W) = P(Y|do(X), Z)$
- Z blocking all back-door paths: $P(Y|do(X), Z) = P(Y|X, Z)$
- no causal paths from X to Y: $P(Y|do(X)) = P(Y)$

Counterfactuals

so far: average causal effects over (sub-)populations by means of interventions (RCTs, do-calculus)

individual causal effects, acting on individual units u, as counterfactuals:

What happened? and *What could have happened?*

only one realization per individual: cannot simply measure difference between respective potential outcomes (all but one purely hypothetical)

counterfactual notation (had X been x): $Y_{X=x}(u)$

Estimation of Individual Causal Effects

impossible to correctly estimate individual causal effects by means of statistical interpolation techniques (impute missing data of potential outcomes)

→ need for additional causal assumptions (a causal model)

two different approaches:

1. structural causal models (Pearl): guided by causal graph and response functions
2. Rubin causal model (potential outcomes): guided by structural assumptions (e.g., the need for adjusting for all confounders)

Structural Causal Models (SCM)

also known as structural equation models (SEM)

modeler needs to specify which (causal graph) and how (deterministically, e.g., linearly) variables interact

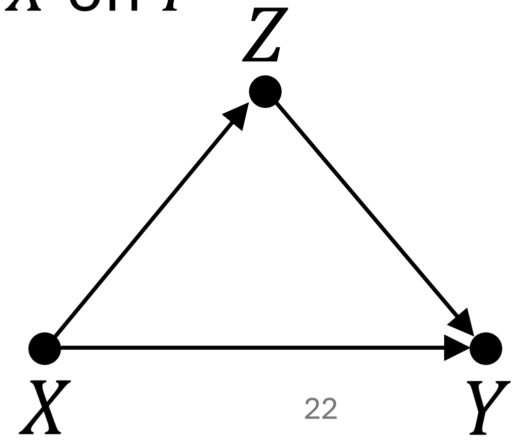
→ average and individual causal effects can be inferred from the model

example: direct and indirect (via mediator Z) causal effects of X on Y

$$Y(u) = f_Y(X(u), Z(u), U_Y(u))$$

$$Z(u) = f_Z(X(u), U_Z(u))$$

U: effects from unobserved (exogenous) variables



Counterfactual Queries in SCM

receipt for counterfactual queries on individual u:

1. abduction: use data to estimate individual $U(u)$ (potentially ML)
2. action: do-operation to change model according to counterfactual query (erasing arrows)
3. prediction: use modified model and updated information on $U(u)$ to estimate individual causal effect

connection to generative models:

unobserved variables U in SCMs correspond to latent noise variables in generative models

→ both use reparametrization trick: randomness as exogenous model input (rather than intrinsic component)

Potential Outcomes with ML

deconfounding

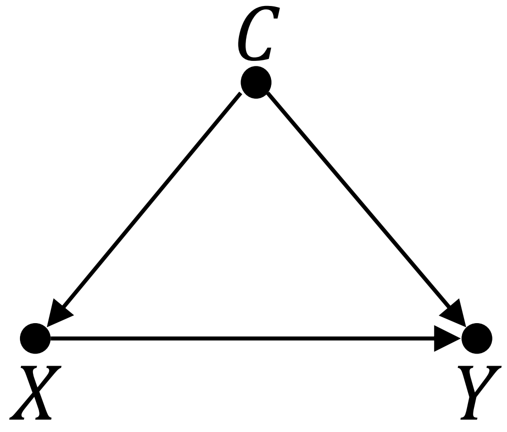
by RCTs or

independence weights (inverse propensity scores)

prediction of individual causal effect with ML model

(generalization by function approximation)

Learned Independence Weights



scenario: X binary (for simplicity), several confounders C

aim: prediction of individual causal effect using observational data only

i. ML model to predict past action policy $P(X|C)$ (beware: need to include all C)

ii. inverse propensity score weighting of each unit (to adjust for multiple confounders):

$$P(Y|do(X)) \propto \sum_{c} \frac{P(Y|X, C=c)}{P(X|C=c)}$$

iii. train ML model on deconfounded data to predict Y with X and C as features

iv. individual causal effect as difference between predictions for setting feature $x = 1$ and $x = 0$, respectively (what-if scenario)

Causality and ML

causation as addition to ML: synonym for understanding

- no ML method can understand causes and effects from data (statistical dependencies) alone → need for causal assumptions (causal model)
- examples: transportability and robustness of models (generalization), handling of missing data, causal explainability

(see Pearl and Schölkopf 1 2 3)

but also: ML as help for causal discovery and inference

- counterfactuls (potential outcomes) need matching of individual cases
- adjusting for many confounders (e.g., propensity scores)